

1. HALAMAN JUDUL

PitStop MVP Product & Technical Specification v1.0

Status: Ready for Development

Referensi desain: PitStop Design v1.0

Tanggal: Jul 22, 2026

Jenis dokumen: Product and Technical Specification

Versi dokumen: 1.0

Tujuan — Pedoman final untuk product, design implementation, engineering, security, QA, dan delivery MVP PitStop.

2. INFORMASI DOKUMEN

Atribut	Nilai
Nama dokumen	PitStop MVP Product & Technical Specification v1.0
Versi	1.0
Status	Ready for Development
Pemilik	Product, Engineering, Design, QA PitStop
Referensi Figma	PitStop Mobile PWA & Admin MVP · PitStop Design v1.0
Tanggal dibuat	Jul 22, 2026
Tanggal diperbarui	Jul 22, 2026
Audience	PM, Designer, Tech Lead, FE, BE, QA, DevOps, Admin Operations
Source of truth	Dokumen: logika produk/teknis; Figma pages utama: visual/interaksi; Prototype Flow: urutan interaksi saja.

3. RIWAYAT REVISI

Versi	Tanggal	Perubahan	Status
1.0	Jul 22, 2026	Spesifikasi final MVP selaras dengan desain final.	Ready for Development

4. DAFTAR ISI

Daftar isi otomatis akan diperbarui saat dokumen dibuka di Google Docs.

5. EXECUTIVE SUMMARY

PitStop adalah Progressive Web App mobile-first bagi driver ojol yang perlu makan, ngopi, toilet, musala, atau istirahat ketika bekerja. Fokus utama MVP adalah menemukan tempat makan murah terdekat yang memiliki menu utama sesuai budget. Guest dapat mencari, memilih lokasi otomatis atau manual, melihat hasil, detail, peta, dan membuka Google Maps tanpa akun. Login hanya dipakai untuk aksi yang mengubah atau melacak data.

MVP memakai Turborepo modular monolith: Next.js App Router untuk web/admin, NestJS dengan Fastify untuk REST API, MySQL 8.4 Spatial, Drizzle, Redis/BullMQ worker, serta object

storage S3-compatible. Semua contribution aplikasi dan Google Form masuk moderation queue sebelum publikasi.

6. LATAR BELAKANG DAN MASALAH

Driver ojol bekerja dengan waktu singkat, uang pegangan terbatas, perangkat kecil, dan jaringan yang tidak selalu stabil. Peta umum tidak menjawab apakah tempat memiliki menu utama murah, sedang buka, dekat, mempunyai fasilitas yang dibutuhkan, atau datanya masih valid. PitStop mengurangi keputusan menjadi kebutuhan + budget; sistem menangani filter, jarak, status buka, ranking, dan data terverifikasi.

7. VISI, TUJUAN, DAN NILAI PRODUK

Prinsip	Keputusan produk
Guest First	Search, lokasi, detail, peta, dan Arahkan tanpa login.
Needs First	Home dimulai dari kebutuhan; Makan Murah paling menonjol.
Budget First	Harga menu utama adalah hard filter Makan Murah.
Recommendation First	Satu primary, maksimal tiga alternatif; map secondary.
Verified Data	Contribution/report dimoderasi; status data terlihat.
Safe While Driving	Kontrol besar, teks ringkas, tidak meminta interaksi saat berkendara.
Progressive Enhancement	Fallback untuk lokasi, jaringan, dan data kosong.

Visi: utilitas singgah yang dapat diandalkan di area kerja padat. Indikator awal: Home ke Arahkan dalam maksimal tiga interaksi, primary cocok budget, contribution dimoderasi end-to-end, dan state kegagalan mudah dipahami.

8. TARGET PENGGUNA

Persona	Tujuan	Kebutuhan/limit
Driver ojol aktif	Menemukan singgahan cepat.	Jarak, status buka, menu utama, fasilitas; satu tangan, koneksi fluktuatif.
Guest	Mencari tanpa komitmen akun.	Tidak diblokir login; lokasi manual bila GPS ditolak.
Kontributor	Menambah/memperbaiki data.	Status contribution, alasan revisi, login saat submit.
Admin/moderator	Menjaga kualitas publik.	Queue, geocoding, pin correction, duplicate check, audit.

9. SCOPE MVP

In scope	Out of scope	Future consideration
PWA; location; manual area; category/budget/recommendation/radius; detail/map/navigation; contribution; Google Form; Google/email auth; activity; reports; admin; worker; audit; offline state.	Pembayaran, dompet, voucher, poin, rating, komentar, chat, forum, iklan, ML recommendation, navigasi internal, native iOS/Android, microservices, GraphQL, Elasticsearch, Kafka, Kubernetes.	Ekspansi wilayah, geocoder produksi, CSV terkontrol, notifikasi tambahan, dan quality automation setelah MVP stabil.

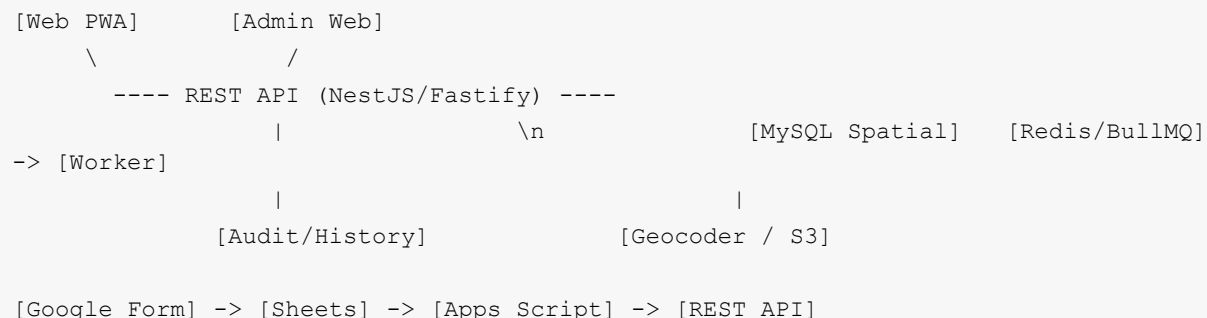
10. ASUMSI DAN BATASAN

- Pengguna memakai smartphone; jaringan dapat tidak stabil.
- Izin lokasi bisa ditolak; manual location wajib tersedia.
- Data tempat bisa tidak lengkap; foto opsional.
- Data publik harus diverifikasi admin.
- Google Maps dipakai untuk navigasi eksternal.
- Pilot wilayah dapat dibatasi demi kualitas data.

11. INFORMASI ARSITEKTUR PRODUK

Empat aplikasi berada dalam monorepo: apps/web, apps/admin, apps/api, apps/worker. API memiliki aturan bisnis serta akses database; worker menangani pekerjaan lambat/retryable. Leaflet/OpenStreetMap digunakan untuk peta, provider geocoding diabstraksikan, dan Google Maps deep link untuk navigasi.

Diagram / Contoh — Arsitektur tingkat tinggi



12. ARCHITECTURE DECISION RECORD

ADR-001 Modular Monolith

Context: domain MVP masih berkembang. Options: microservices atau modular monolith.

Decision: modular monolith. Rationale: delivery cepat dan transaction lintas domain sederhana.

Consequence/review: dependency module ketat; review jika module perlu deploy/scale mandiri.

ADR-002 REST API

Context: web, admin, Apps Script, worker butuh kontrak umum. Options: GraphQL atau REST.

Decision: REST /api/v1 + OpenAPI. Rationale: cocok resource, caching, observability. Review bila query agregat tidak efisien.

ADR-003 Next.js untuk Web dan Admin

Context: UX driver dan moderator berbeda. Options: SPA tunggal atau Next.js dua app.

Decision: apps/web dan apps/admin Next.js App Router. Consequence: shared package wajib menjaga konsistensi.

ADR-004 NestJS dengan Fastify

Context: API perlu module, DI, performa HTTP. Options: Express atau Fastify. Decision:

NestJS/Fastify. Consequence: plugin Express tidak diandalkan.

ADR-005 MySQL Spatial

Context: radius/jarak adalah inti. Options: PostgreSQL/PostGIS atau MySQL Spatial. Decision: MySQL 8.4 POINT SRID 4326/SPATIAL INDEX. Review bila analitik spatial menjadi kompleks.

ADR-006 Drizzle ORM

Context: TypeScript strict dan SQL spatial eksplisit. Options: Prisma atau Drizzle. Decision:

Drizzle + mysql2. Consequence: migration/query raw ditinjau ketat.

ADR-007 Redis dan BullMQ

Context: geocoding/integrasi tidak boleh memblokir request. Decision: Redis/BullMQ dedicated worker. Consequence: Redis health, retry, DLQ, replay wajib.

ADR-008 Guest First Authentication

Context: search tidak boleh berfiksi. Decision: login hanya mutation/activity. Consequence: abuse mitigation di endpoint mutation.

ADR-009 Geocoding Provider Abstraction

Context: dev/prod provider bisa berbeda. Decision: GeocodingPort dengan adapter. Consequence: cache, rate limit, confidence seragam.

ADR-010 MinIO dan S3-Compatible Storage

Context: foto opsional memerlukan local/prod parity. Decision: MinIO dev, S3-compatible prod. Consequence: presigned URL/lifecycle/validation wajib.

ADR-011 PWA

Context: mobile web dan jaringan tidak stabil. Decision: Serwist PWA. Consequence: cache aman dan offline fallback; token tidak di-cache.

ADR-012 Separate Web and Admin Apps

Context: user dan admin berbeda. Decision: dua app. Consequence: role isolation dan config shared package.

13. INFORMATION ARCHITECTURE

Area	Halaman
Pengguna	Home; Recommendations; Place Detail; Map Results; Manual Location.
Contribution	Add Place Step 1, Step 2, Step 3; Guest Login; Email Login; Contribution Success; Contribution Detail.
Aktivitas	Activity; Activity Guest; Report Change.
Admin	Dashboard; Moderation; Decision Revision/Reject/Approve; Places/Places Edit; Reports/Apply/Reject.

Visual source of truth adalah Cover & Design System, Components, Mobile Core Screens, Mobile States & Overlays, Mobile Contribution Flow, Admin Dashboard, dan Admin Moderation Flow. Prototype Flow hanya untuk relasi layar.

14. USER FLOWS

Flow	Actor	Precondition	Main flow	Alternative/failure	Postcondition
Cari makan sesuai budget	Guest	Home + lokasi	Pilih budget → Cari → Loading → Recommendations → Detail → Arahkan	Network error/ubah lokasi	Google Maps terbuka
Hasil di luar radius	Guest	Tidak ada 0–5 km	Out of Radius → Lihat tempat 5–10 km	Ubah pencarian	Fallback dipahami
Harga di atas budget	Guest	Tempat dekat > budget	Above Budget → Tetap lihat / Ubah budget	No budget match	Tidak diklaim sesuai
Lokasi belum aktif	Guest	GPS ditolak	Gunakan GPS lagi / pilih area manual	GPS gagal	Home memakai area
Peta hasil	Guest	Recommendations ada	Lihat peta → pin → Detail/Arahkan	Map error → list	List tetap usable
Tambah tempat sebagai guest	Guest→User	Step data terisi	Step 1→2→3→login→success	Draft/validation	PENDING dibuat
Aktivitas/revisi	User	Needs revision	Activity→detail→perbaiki→submit	Guest state	Riwayat tampil
Moderasi admin	Admin	Contribution queue	cek→revision/reject/approve	geocode gagal	audit/status update
Pilih lokasi manual	Guest	GPS unavailable	pilih Tambora→Gunakan area	GPS retry	context area tersimpan
Login email	Guest	Guest Login	email/password→Masuk	input error	session aman
Kelola tempat	Admin	Place published	Places→Edit→Simpan	status nonaktif	history/audit
Admin reports	Admin	Report pending	detail→Apply/Reject	bukti kurang	report+place update

15. BUSINESS RULES

Domain	Aturan
Kategori	Hanya Makan Murah, Ngopi, Toilet, Musala, Istirahat; Makan Murah utama.

Domain	Aturan
Place	DRAFT, PENDING, ACTIVE, TEMPORARILY_CLOSED, PERMANENTLY_CLOSED, ARCHIVED; tidak hard delete.
Contribution	DRAFT, PENDING, IN_REVIEW, NEEDS_REVISION, APPROVED, REJECTED, MERGED; semua sumber satu queue.
Report	PENDING, IN_REVIEW, APPLIED, REJECTED; apply mencatat history/audit.
Verification	UNVERIFIED, ADMIN_VERIFIED, COMMUNITY_CONFIRMED, STALE; primary harus ADMIN_VERIFIED.
Facility	AVAILABLE, NOT_AVAILABLE, UNKNOWN; UNKNOWN bukan AVAILABLE.
Login/publish	Guest search; user untuk mutation; approve memakai transaction+audit.

16. ATURAN BUDGET

Pilihan: $\leq$ Rp10.000, $\leq$ Rp15.000, $\leq$ Rp20.000, $\leq$ Rp25.000, Bebas, dan manual rupiah positif. Budget terakhir disimpan lokal. Makan Murah sesuai jika terdapat minimal satu menu utama valid dengan $\text{price_amount} \leq \text{budget}$. Harga minuman atau gorengan satuan tidak boleh menjadi indikator menu utama. price_from adalah menu utama valid termurah; state Above Budget muncul jika kandidat dekat lebih mahal.

17. ATURAN REKOMENDASI

Pipeline: ACTIVE $\rightarrow$ ADMIN_VERIFIED $\rightarrow$ coordinate valid $\rightarrow$ kategori $\rightarrow$ menu utama valid $\rightarrow$ budget hard filter $\rightarrow$ open status $\rightarrow$ distance $\rightarrow$ score $\rightarrow$ one primary + max three alternative. $\text{Score} = 0.35 \text{ budgetFit} + 0.30 \text{ distanceScore} + 0.20 \text{ openScore} + 0.10 \text{ freshnessScore} + 0.05 \text{ communityScore}$; semua 0–1. $\text{distanceScore} = \max(0, 1 - \text{distanceKm} / \text{radiusKm})$; open score Buka=1, Segera tutup=.55, Tidak diketahui=.35, Tutup=0.

Diagram / Contoh — Pseudocode recommendation

```

candidates = active && adminVerified && validCoordinate && categoryMatch
if category == MAKAN_MURAH: candidates = candidates.filter(validMainMealPrice <=
budget)
ranked = candidates.map(openStatus, distance, score).filter(notClosed).sort(desc
score)
return primary=ranked[0], alternatives=ranked[1:4], reasons=maxThree

```


18. ATURAN RADIUS

Band/state	Perilaku
0–5 km	Primary recommendation.
>5–10 km	Fallback pertama dengan jarak aktual.
>10–20 km	Fallback kedua atas pilihan user.
>20 km	Tidak otomatis ditampilkan.
No verified data	CTA cari lebih jauh/tambah tempat.
No budget match	Jelaskan tidak ada menu utama valid.
All places closed	Tawarkan tempat buka lebih jauh/lihat semua.
Location denied	GPS retry atau manual area.

19. ATURAN STATUS BUKA

Timezone Asia/Jakarta. Buka bila kini berada interval `operating_hours`; Segera tutup bila sisa ≤ 60 menit; Tutup bila luar interval; Tidak diketahui bila data kosong/tidak valid. Jadwal 20.00–02.00 melintasi tengah malam. Hari libur/exception menimpa jadwal rutin. Tempat Tutup tidak menjadi primary kecuali user melihat semua.

20. DESIGN SYSTEM IMPLEMENTATION

Token	Nilai
Primary Green	#16A34A
Primary Green Dark	#166534
Night Navy	#0F2742
Accent Orange	#F97316
Background	#F8FAFC
Surface	#FFFFFF
Text Primary	#0F172A
Text Secondary	#64748B
Border	#E2E8F0
Error	#DC2626
Warning Background	#FFF7ED
Success Background	#F0FDF4

Inter; spacing 4/8 px; radius card 18 px, button/input 12 px; touch target minimal 48×48; metadata minimal 13 px. Map Figma component ke packages/ui: Button, FormInput, StatusBadge, FacilityChip, PlaceCard, BottomSheet, Modal, EmptyState, LoadingSkeleton, AdminSidebar, AdminStatCard, ContributionListItem. Tailwind memakai token semantic, Radix primitive, CVA variant.

21. SYSTEM ARCHITECTURE

Module boundary mengikuti domain. Controller validasi request; application service memegang policy/transaction; repository Drizzle mengakses MySQL; worker memanggil service aman. API tidak geocode synchronous dalam public request.

Diagram / Contoh — Request/job flow

```
Browser Web/Admin -> Next.js -> REST /api/v1 -> NestJS Modules -> MySQL
                                     |                               \n
v                               Redis Queue -> Worker -> providers
                                     OpenAPI contract + requestId
```

22. MONOREPO STRUCTURE

Diagram / Contoh — Turborepo layout

```
apps/web           PWA pengguna
apps/admin         dashboard desktop
apps/api           NestJS Fastify REST
apps/worker        BullMQ consumer
packages/ui        database contracts validation config testing
infrastructure/docker mysql redis minio monitoring
docs/adr api database product
```

Tidak ada akses database dari web/admin. packages/database menyimpan Drizzle schema/migration/seed; contracts menyimpan types DTO; validation menyimpan Zod; config menangani environment.

23. BACKEND MODULES

Module	Responsibility	Permission	Job
auth	OAuth/email/session	public+user	cleanup sessions
users	profile/roles	user/admin	—
places	public place/status	public read/admin write	stale refresh

Module	Responsibility	Permission	Job
categories	master categories	public read/admin	—
menus	main meal/price	admin	—
recommendations	filter/rank	public	—
geolocation	spatial/geocode	admin verify	geocode
contributions	draft/submit/revision	owner/admin	notify
moderation	decision/merge	admin	duplicate detect
reports	change reports	user/admin	notify
confirmations	community confirmation	user	—
integrations	Google Form inbound	signed	form process
uploads	presigned/media	scoped	optimize
notifications	status events	internal	send
audit-logs	append-only log	system/admin	—
health	health/metrics	internal	—

24. DATABASE DESIGN

ULID CHAR(26) menjadi identifier API/database yang sortable. created_at/updated_at UTC ada di tabel mutable; deleted_at dipakai jika soft delete relevan. POINT selalu POINT(longitude, latitude) SRID 4326; SPATIAL INDEX pada places.location. Gunakan ST_Distance_Sphere dengan parameter, bukan string SQL interpolation.

Diagram / Contoh — Konvensi MySQL spatial

```
location POINT NOT NULL SRID 4326
SPATIAL INDEX idx_places_location (location)
ST_Distance_Sphere(location, ST_SRID(POINT(?, ?), 4326))
-- parameter order: longitude, latitude
```

Tabel	Tujuan	Kolom inti	Constraint/index/nullability
users	identity	id,email,display_name,status	PK ULID; email unique; deleted_at
roles	role master	id,code	PK; code unique
user_roles	user-role	user_id,role_id	PK gabungan; FK
auth_accounts	OAuth link	id,user_id,provider,provider_account_id	provider/account unique

Tabel	Tujuan	Kolom inti	Constraint/index/nullability
refresh_tokens	session	id,user_id,token_hash,expires_at,revoked_at	user/expires index
places	place public	id,name,address,location,status,verification	SPATIAL location; status index; soft delete
categories	MVP categories	id,code,name	code unique
place_categories	relation	place_id,category_id,is_primary	unique pair
menus	price	id,place_id,name,price_amount,is_main_meal	place/price index; positive check
facilities	facility master	id,code,name	code unique
place_facilities	availability	place_id,facility_id,availability	unique pair
operating_hours	hours	id,place_id,weekday,opens_at,closes_at	place/day unique
place_photos	place media	id,place_id,object_key,mime_type,status	place/status index
contributions	submission	id,user_id,source_id,status,place_id	status/time index
contribution_payloads	canonical data	id,contribution_id,payload_json	contribution unique
contribution_photos	submission media	id,contribution_id,object_key,status	contribution index
moderation_reviews	decision	id,contribution_id,admin_id,decision,notes	contribution/time index
place_confirmations	community	id,place_id,user_id,status	dedupe unique
place_reports	report	id,place_id,user_id,type,status,suggested_json	status/time index
report_evidence	report media	id,report_id,object_key,mime_type	report index
integration_sources	integration	id,code,secret_hash,active	code unique
google_form_submissions	inbox	id,source_id,external_submission_id,payload_json,status	source+external unique
geocoding_results	geocode result	id,subject,provider,confidence,raw_json,status	subject/provider index
audit_logs	audit	id,actor_id,action,entity_type,entity_id,before/after	entity/time index; append-only
place_change_history	place history	id,place_id,source,change_json,applied_at	place/time index

25. DATABASE TRANSACTIONS

Operation	Urutan	Safety
Approve contribution	Lock contribution → validate status → create/update place → menus/facilities/hours → audit/history → APPROVED → commit	rollback penuh; key contribution mencegah double approve
Merge duplicate	Lock source/target → merge chosen data → history/audit → MERGED	source tetap terlacak
Apply report	Lock report/place → validate → update place → history/audit → APPLIED	optimistic version
Close place	Lock → status TEMPORARY/PERMANENT → audit	no hard delete
Update place	Lock → validate → update related rows → history	conflict 409
Revoke refresh	Lock → revoked_at → audit if admin	idempotent

26. API DESIGN

Prefix /api/v1. Response sukses: {success:true,data,meta?}; mutation menggunakan Idempotency-Key kecuali refresh/logout. Semua route punya requestId, OpenAPI, Zod/DTO, rate limit, dan audit sesuai tabel.

Method	Path	Auth	Purpose/input
GET	/categories	public	kategori
GET	/places/recommendations	public	category,budget,lat/lng/area,radius
GET	/places	public	filters/page
GET	/places/:id	public	detail
GET	/places/map	public	bbox/category/budget
POST	/auth/register	public	email,password
POST	/auth/login	public	email,password
POST	/auth/google	public	OAuth code/PKCE
POST	/auth/refresh	cookie	rotation
POST	/auth/logout	user	revoke
POST	/contributions	user	draft payload

Method	Path	Auth	Purpose/input
PATCH	/contributions/:id	owner	update draft
POST	/contributions/:id/submit	owner	submit
GET	/contributions/me	user	activity
GET	/contributions/:id	owner/admin	detail
POST	/places/:id/confirmations	user	confirmation
POST	/places/:id/reports	user	report
GET	/admin/dashboard	admin	queue stats
GET	/admin/contributions	admin	filters
GET	/admin/contributions/:id	admin	detail
POST	/admin/contributions/:id/approve	admin	resolution
POST	/admin/contributions/:id/reject	admin	reason
POST	/admin/contributions/:id/request-revision	admin	fields/reason
POST	/admin/contributions/:id/merge	admin	targetPlaceId
GET	/admin/places	admin	filters
PATCH	/admin/places/:id	admin	edit/status
GET	/admin/reports	admin	filters
POST	/admin/reports/:id/apply	admin	resolution
POST	/admin/reports/:id/reject	admin	reason
POST	/integrations/google-form/places	signed integration	submission/signature

Diagram / Contoh — Request/response example

```
POST /api/v1/contributions
Idempotency-Key: 01J...
{ "name": "Warung Bu Ani", "categories": ["MAKAN_MURAH"],
  "mainMenu": {"name": "Nasi telur", "priceAmount": 12000},
  "address": "Tambora, Jakarta Barat" }
201 {"success": true, "data": {"id": "01J...", "status": "DRAFT"}}
```

27. API ERROR FORMAT

```
{ "success": false, "error": { "code": "PLACE_NOT_FOUND",
```

```
"message":"Tempat tidak ditemukan.", "details":{}, "requestId":"req_01J..." }
```

Code	HTTP	Meaning
VALIDATION_ERROR	400	Input/query invalid
UNAUTHENTICATED	401	Session/token invalid
FORBIDDEN	403	Role/ownership missing
PLACE_NOT_FOUND	404	Place absent
CONTRIBUTION_INVALID_STATE	409	State conflict
DUPLICATE_SUBMISSION	409	Idempotency duplicate
LOCATION_REQUIRED	422	Location/area required
NO_BUDGET_MATCH	422	No valid main menu match
RATE_LIMITED	429	Limit exceeded
INTEGRATION_SIGNATURE_INVALID	401	Signature invalid
GEOCODING_UNAVAILABLE	503	Provider unavailable
INTERNAL_ERROR	500	Unexpected; requestId

28. AUTHENTICATION AND AUTHORIZATION

Email/password memakai Argon2id. Access token pendek (target 15 menit); refresh opaque/hash dalam HttpOnly Secure SameSite=Lax cookie, rotation setiap refresh, session revocation. Login dibatasi per IP+identifier, error generik untuk cegah enumeration, password reset token sekali pakai berumur pendek.

Capability	Guest	User	Admin
Search/location/detail/map/navigation	Yes	Yes	Yes
Contribution/report/confirmation	No	Own	Yes
Activity/revision	No	Own	Yes
Moderation/places/admin reports	No	No	Yes
Audit log	No	No	Policy-scoped

29. GOOGLE FORM INTEGRATION

Google Form → Google Sheets → Apps Script → POST integration → google_form_submissions → contribution PENDING → geocoding/dedupe worker → moderation. Form tidak meminta latitude/longitude.

Field	Policy
Nama tempat/alamat/wilayah/kategori	Wajib
Patokan/Maps link	Opsional
Menu utama/harga/kisaran/budget	Wajib untuk Makan Murah; integer rupiah
Fasilitas/jam/catatan/email	Optional/validated

Apps Script mengirim source, submission ID unik, timestamp UTC, canonical payload, HMAC signature/integration key. API memverifikasi signature dan memakai UNIQUE(source_id, external_submission_id). 202 berarti accepted; retry tidak membuat contribution kedua. Dashboard menampilkan sync status.

30. GEOCODING

Diagram / Contoh — TypeScript interface

```
interface GeocodingPort {  
  geocode(input: {address:string;area?:string;mapUrl?:string}):Promise<Result>;  
  reverse(point:{longitude:number;latitude:number}):Promise<Result>;  
}  
Result={provider,longitude?,latitude?,confidence,raw,status}
```

Nominatim untuk development dengan rate limit/cache/retry. Hasil confidence rendah/gagal tidak otomatis publish. Admin dapat cari alamat, geser pin, dan verify lokasi; raw response dibatasi retention.

31. BACKGROUND JOBS

Job	Trigger/payload	Retry/failure
geocode-contribution	submitted contribution	5 exponential; 30s; DLQ
reverse-geocode-place	pin correction	3 retry; cache
detect-duplicate-place	geocode success	name+distance; human decision
process-google-form-submission	inbox accepted	idempotent; retry 5
optimize-place-photo	upload accepted	scan/resize; quarantine

Job	Trigger/payload	Retry/failure
notify-contribution-status	status transition	retry 3; non-blocking
refresh-stale-place-data	scheduled cursor	batch idempotent
cleanup-expired-sessions	scheduled timestamp	safe revoke/delete

Setiap job membawa request/correlation ID, idempotency key, Pino log, timeout, attempt counter, error class, dan DLQ/replay.

32. FILE STORAGE

MinIO development, S3-compatible production. Key: {env}/{entity}/{ulid}/{original|derivative}; upload presigned URL scoped. Allowlist JPEG/PNG/WebP, max 5 MB, MIME sniffing, dimensions, checksum optional. Worker scan/optimise/thumbnail. Soft delete diikuti grace period dan lifecycle deletion sesuai retention.

33. PWA DAN OFFLINE

Manifest+Serwist membuat aplikasi installable. Cache shell statis; stale-while-revalidate untuk kategori dan hasil terakhir yang aman. Offline fallback menjelaskan status dan dapat menampilkan hasil terakhir. Tidak cache access token, refresh response, admin route, presigned URL, atau data personal/moderation. Mutation tidak dianggap terkirim sebelum server ack; draft lokal boleh disimpan tanpa token sensitif.

34. SECURITY REQUIREMENTS

- Zod/DTO allowlist, Drizzle parameterization, raw SQL parameterized.
- Output encode/CSP, CSRF defense cookie mutation, CORS allowlist, secure headers.
- RBAC/ownership guard; admin route ADMIN only; audit decision.
- Rate limits login/search/integration/mutation; Pino redaction password/token/cookie/signature.
- Upload MIME/size/object-key validation and quarantine.
- Secret manager/environment; HTTPS production; OWASP review.

35. PRIVACY DAN DATA RETENTION

Data	Use	Initial retention
User profile	auth/activity	active account + 30d deletion request
Query location	runtime recommendation	no precision history; short safe cache

Data	Use	Initial retention
Contribution/report	moderation/audit	24 months after decision
Refresh token	session security	expiry/revocation + 30d
Audit log	governance	24 months restricted
App logs	observability	30–90d redacted
Photos/evidence	moderation	status/lifecycle policy

Deletion request revokes sessions and removes/anonymizes personal data where retention tidak diperlukan; audit residual diminimalkan. Lokasi tidak dipakai untuk profiling atau iklan.

36. OBSERVABILITY

Pino structured log: timestamp, level, requestId, route, status, latency, actor role redacted. API: /health/live, /health/ready, /metrics. Monitor queue depth/failed jobs, geocoding, API error/latency, DB connections, storage. OpenTelemetry ditambahkan bertahap untuk trace request→service→job. Audit log berbeda dari application log.

37. TESTING STRATEGY

Layer	Tool	Key coverage
Unit	Vitest	scoring,budget,open status,guard
Integration	Vitest+Testcontainers	MySQL spatial, Redis queue, transaction
API	Supertest	contract,auth,error,idempotency
E2E	Playwright	guest,location,contribution,admin
Accessibility	axe-core+manual	WCAG/focus/labels
Load	k6	recommendation/map/login
Security smoke	OWASP checklist	auth,CSRF,upload,RBAC

Critical cases: POINT lng/lat, valid main menu only, overnight hours, job retry, approval rollback, report history, refresh rotation, Google Form duplication, and guest login boundary.

38. ACCESSIBILITY

Target WCAG 2.2 AA. Tap target $\geq 48 \times 48$, metadata $\geq 13\text{px}$, contrast token, visible focus, semantic HTML, aria-describedby form errors, keyboard navigation, reduced motion, and list/navigasi alternative untuk map. Jangan jadikan pin map satu-satunya pembawa informasi.

39. PERFORMANCE TARGET

Metric	Dev	Prod
LCP P75	≤3.0s	≤2.5s
CLS	≤0.10	≤0.05
INP P75	≤250ms	≤200ms
API P95	≤500ms	≤350ms
Recommendation P95	≤700ms	≤450ms
Map load	≤3.5s	≤2.5s
Initial JS	≤250KB gzip	≤200KB gzip
Optimised image	≤500KB upload	≤250KB delivery
PWA	manifest/SW valid	install eligible

40. DEVELOPMENT ENVIRONMENT

WSL 2, Node LTS, pnpm, Docker Compose, MySQL, Redis, MinIO, Mailpit. Example local ports: web 3000, admin 3001, API 3002, MySQL 3306, Redis 6379, MinIO 9000, Mailpit 8025. packages/config validates .env; no secret in repository.

Diagram / Contoh — Environment example

```
DATABASE_URL=mysql://pitstop:pitstop@localhost:3306/pitstop
REDIS_URL=redis://localhost:6379
S3_ENDPOINT=http://localhost:9000
S3_BUCKET=pitstop-local
GOOGLE_OAUTH_CLIENT_ID=...
INTEGRATION_HMAC_SECRET=...
```

41. CI/CD

Diagram / Contoh — GitHub Actions pipeline

```
Install → Lint → Typecheck → Unit Test → Integration Test → Build
      → E2E Smoke → Docker Build → Security Scan
```

Protected main, short-lived branches, PR review, migration review, green gates, no secrets in logs, dependency/container scan before promotion.

42. DEPLOYMENT ARCHITECTURE

Deploy web, admin, API, worker sebagai service/container independen; MySQL/Redis managed; S3-compatible storage; secrets/environment terpisah dev/staging/prod. Migration backward-compatible dijalankan terkontrol sebelum rollout. Backup MySQL diuji restore; Redis bersifat queue/cache dengan replay/DLQ. Vendor tidak dikunci.

43. SEED DATA

Data berikut simulasi untuk development/test, bukan data publik produksi.

Nama	Data simulasi
Warung Bu Ani	Kategori: Makan Murah Menu utama/harga: Nasi telur Rp12.000 Area & koordinat simulasi: Tambora · 106.8061, -6.1468 Fasilitas: Toilet, Parkir · Jam: 08.00–22.00 Status/verifikasi: ACTIVE · ADMIN_VERIFIED
Warkop Bang Udin	Kategori: Ngopi Menu/harga: Kopi Rp5.000 Area & koordinat simulasi: Tambora · 106.8039, -6.1491 Fasilitas: Colokan, Wi-Fi · Jam: 17.00–01.00 Status/verifikasi: ACTIVE · ADMIN_VERIFIED
Warteg Barokah	Kategori: Makan Murah Menu utama/harga: Nasi rames Rp13.000 Area & koordinat simulasi: Grogol · 106.7972, -6.1670 Fasilitas: Parkir, Toilet · Jam: 07.00–21.00 Status/verifikasi: TEMPORARILY_CLOSED · ADMIN_VERIFIED
Nasi Uduk Ibu Rini	Kategori: Makan Murah Menu utama/harga: Nasi uduk telur Rp14.000 Area & koordinat simulasi: Cengkareng · 106.7340, -6.1430 Fasilitas: Area teduh · Jam: 05.30–11.00 Status/verifikasi: ACTIVE · COMMUNITY_CONFIRMED
Warung Madura 24 Jam	Kategori: Istirahat Menu/harga: Mie instan Rp10.000 Area & koordinat simulasi: Kalideres · 106.7030, -6.1380 Fasilitas: Parkir, Colokan · Jam: 00.00–24.00 Status/verifikasi: ACTIVE · ADMIN_VERIFIED

44. ACCEPTANCE CRITERIA

ID	Given / When / Then
AC-01	Given the relevant precondition, when the user/system performs guest search tanpa login, then the documented result, error handling, authorization, and audit behavior are observable and testable.

ID	Given / When / Then
AC-02	Given the relevant precondition, when the user/system performs gps update lokasi, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-03	Given the relevant precondition, when the user/system performs manual tambora update home, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-04	Given the relevant precondition, when the user/system performs budget hard filter main meal, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-05	Given the relevant precondition, when the user/system performs one primary/max three alternative, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-06	Given the relevant precondition, when the user/system performs outside radius state+actual distance, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-07	Given the relevant precondition, when the user/system performs no verified data cta, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-08	Given the relevant precondition, when the user/system performs all places closed state, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-09	Given the relevant precondition, when the user/system performs detail menu/facility/hours/verification, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-10	Given the relevant precondition, when the user/system performs map pin+detail/arahkan, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-11	Given the relevant precondition, when the user/system performs google maps deep link, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-12	Given the relevant precondition, when the user/system performs contribution submit pending, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-13	Given the relevant precondition, when the user/system performs guest login only at submit, then the documented result, error handling, authorization, and audit behavior are observable and testable.

ID	Given / When / Then
AC-14	Given the relevant precondition, when the user/system performs activity guest cta login, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-15	Given the relevant precondition, when the user/system performs google form dedupe, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-16	Given the relevant precondition, when the user/system performs geocode low confidence requires admin, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-17	Given the relevant precondition, when the user/system performs approve atomic+audit, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-18	Given the relevant precondition, when the user/system performs place close without delete, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-19	Given the relevant precondition, when the user/system performs apply report history, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-20	Given the relevant precondition, when the user/system performs job retry/dlq, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-21	Given the relevant precondition, when the user/system performs error format requestid, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-22	Given the relevant precondition, when the user/system performs point lng/lat/spatial index, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-23	Given the relevant precondition, when the user/system performs pwa offline fallback, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-24	Given the relevant precondition, when the user/system performs wcag touch/focus/contrast, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-25	Given the relevant precondition, when the user/system performs ci gates pass, then the documented result, error handling, authorization, and audit behavior are observable and testable.

ID	Given / When / Then
AC-26	Given the relevant precondition, when the user/system performs admin rbac reject non-admin, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-27	Given the relevant precondition, when the user/system performs ui matches figma source pages, then the documented result, error handling, authorization, and audit behavior are observable and testable.
AC-28	Given the relevant precondition, when the user/system performs metric/log correlation available, then the documented result, error handling, authorization, and audit behavior are observable and testable.

45. DEFINITION OF DONE

- UI aligns with Figma source-of-truth pages.
- Lint/typecheck/unit/integration/E2E/accessibility pass.
- OpenAPI, migration, seed, error handling, audit, observability, security review complete.
- Acceptance criteria met; code review and rollback/migration plan complete.

46. DEVELOPMENT PHASES

Phase	Goal	Deliverables	Exit
0	Repo/tooling	Turborepo, pnpm, Docker, CI	gates green
1	Database	Drizzle schema/migration/seed	spatial tests
2	Design system	UI package/tokens	a11y primitive
3	Public API	places/recommendations	budget/radius
4	Guest slice	Home→Arahkan	E2E
5	Location/map/state	GPS/manual/fallback	state E2E
6	Authentication	Google/email/session	security tests
7	Contribution	draft/submit/activity	E2E
8	Admin moderation	queue/approve	transaction/audit
9	Form/worker	inbox/jobs	retry/idempotency
10	Reports/audit	apply/reject/history	E2E
11	PWA/quality	SW,k6,axe,metrics	quality gates
12	Deployment	infra/backup/rollout	readiness/rollback

47. RISIKO DAN MITIGASI

Risk	Impact	Mitigation
Data sedikit	empty results	pilot area, Form, contribution CTA
Harga usang	budget mismatch	updated_at, stale job, reports
Geocoding salah	wrong pin	confidence, admin pin verification
Contribution abuse	spam	auth/rate/moderation/audit
Foto ilegal	unsafe media	validation/quarantine/lifecycle
GPS rejected	no context	manual location
OSM limits	geocode delay	cache/throttle/provider port
Redis down	job delay	health/retry/DLQ
Migration failure	outage	compat migration/backup
Admin abuse	public damage	RBAC/audit/revocation
Scope creep	delay	in/out scope/ADR
Overengineering	complexity	modular monolith boundary

48. OPEN QUESTIONS

- Wilayah pilot awal/ekspansi.
- Provider geocoding production serta SLA/biaya.
- Provider hosting, S3-compatible storage, domain.
- Email provider password reset.
- Retention final setelah review legal/operasional.

Bukan open question — Stack, Guest First, MySQL Spatial, Drizzle, Redis/BullMQ, kategori MVP, dan batas scope sudah terkunci.

49. GLOSSARY

Term	Definition
Guest	Pengunjung tanpa akun.
Contribution	Data baru/perbaikan pending moderation.
Moderation	Pemeriksaan admin lokasi/harga/kategori/duplikasi.
Verified	Status pemeriksaan data.
Spatial query	Query radius/jarak geospasial.
SRID	Referensi koordinat; 4326/WGS84.
Geocoding	Alamat → koordinat.

Term	Definition
Background job	Kerja asynchronous queue/worker.
Soft delete	Menonaktifkan tanpa menghapus fisik.
Idempotency	Request berulang menghasilkan satu efek.
PWA	Web installable dengan service worker.
Stale	Data melewati batas kebaruan.

50. APPENDIX

Diagram / Contoh — API response

```
{ "success":true, "data": { "primary": {
  "name":"Warung Bu Ani", "distanceKm":1.2, "openStatus":"OPEN",
  "priceFrom":10000, "reasons":["Dekat","Sesuai budget","Sedang buka"] } } }
```

Diagram / Contoh — BullMQ payload

```
{ "job":"geocode-contribution", "contributionId":"01J...",
  "idempotencyKey":"geocode:01J...", "requestId":"req_01J..." }
```

Diagram / Contoh — Spatial query parameterized

```
SELECT id, ST_Distance_Sphere(location, ST_SRID(POINT(?, ?),4326)) distance_m
FROM places WHERE status='ACTIVE' AND verification_status='ADMIN_VERIFIED'
ORDER BY distance_m LIMIT ?;
-- bindings: longitude, latitude, limit
```

Referensi Figma: PitStop Mobile PWA & Admin MVP. Source visual: Cover & Design System, Components, Mobile Core Screens, Mobile States & Overlays, Mobile Contribution Flow, Admin Dashboard, Admin Moderation Flow. Prototype Flow hanya referensi urutan.

Status enum dan permission matrix

Status enum — PLACE: DRAFT, PENDING, ACTIVE, TEMPORARILY_CLOSED, PERMANENTLY_CLOSED, ARCHIVED. CONTRIBUTION: DRAFT, PENDING, IN_REVIEW, NEEDS_REVISION, APPROVED, REJECTED, MERGED. REPORT: PENDING, IN_REVIEW, APPLIED, REJECTED. VERIFICATION: UNVERIFIED, ADMIN_VERIFIED, COMMUNITY_CONFIRMED, STALE. FACILITY: AVAILABLE, NOT_AVAILABLE, UNKNOWN.

Role	Pencarian	Mutation publik	Admin governance
------	-----------	-----------------	------------------

GUEST	Ya	Tidak	Tidak
USER	Ya	Contribution/report/confir mation sendiri	Tidak
ADMIN	Ya	Sesuai policy	Moderation, places, reports, audit